

Multiple Choice (10 marks)

1. Facial flushing is seen in approximately 50% of East Asians and is thought to be due to a deficiency of?
 - (a) ALDH 1
 - (b) ALDH 2
 - (c) ALDH 3
 - (d) ALDH 4
2. Which of the following is not a zinc-dependent enzyme?
 - (a) Superoxide dismutase
 - (b) Alkaline phosphatase
 - (c) Glutathione peroxidase
 - (d) DNA polymerase
3. Ulcerative Colitis
 - (a) Carries a risk of colonic cancer
 - (b) Can be treated with a LOFFLEX diet
 - (c) Can be caused by small intestinal bacterial overgrowth
 - (d) Can be caused by milk intolerance
4. A deficiency of which vitamin has been associated with enamel defects and increased risk of dental caries?
 - (a) Vitamin A
 - (b) Folic acid
 - (c) Vitamin C
 - (d) Vitamin D
5. Which of the following statements is not true?
 - (a) Vegan diets are likely to be deficient in protein
 - (b) Cereals are deficient in lysine
 - (c) Legumes are deficient in methionine
 - (d) Dairy products are rich source of tryptophan

True / False (10 marks)

Answer True or False

6. True or False: Riboflavin is the primary vitamin that provides a coenzyme for fatty acid synthesis.
7. True or False: Potassium is a dietary component that has been shown to help improve hypertension in humans.
8. True or False: Acarbose is the first-line drug for managing type 2 diabetes in patients with obesity as of 2020.
9. True or False: Epidemiological measures of effect assess the strength of the association between an exposure and an outcome.
10. True or False: Starch mainly consists of amylose, which is a polymer of fructose connected by beta 1,4-linkages.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of ATP as the immediate source of chemical energy in skeletal muscle contraction, including its production, usage, and significance in muscle function.
12. Discuss the effectiveness of various treatments for Binge Eating Disorder (BED) and analyze why benzodiazepines have not shown efficacy in managing this condition.

End of Paper